Foundations of Large Language Models

Abilities and Limitations through a Formal Language Learning Perspective

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https://courses.mpi-sws.org/fllm-ws26/

A Shared Inquiry

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Theoretical

Which languages are learnable by LLMs?

Empirical

Does the prediction hold in practice?

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Theoretical

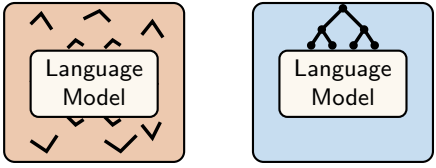
Which languages are learnable by LLMs?

Empirical

Does the prediction hold in practice?

Our perspective is through the lens of **formal language** learning

Why Formal Language?



"An ant, viewed as a behaving system, is quite simple. The apparent complexity of its behavior over time is largely a reflection of the complexity of the environment in which it finds itself."

— Herbert Simon, *The Sciences of the Artificial*, 1996

Advantages of Formal Language

A formal language is a set of strings generated by a formal grammar (representation)

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- **Precise specification.** We can decide whether a string belongs to the language
- **Exact sampling.** Train and test strings are sampled i.i.d.
- **No semantic ambiguity, no contamination.** Pure syntax, and unseen in pretraining
- **Control over structure.** The grammar is a knob on complexity

Our Pursuit, in Three Axes

1

No assumption about the computational model

Which languages are learnable at all?

2

A simplified model of the transformer

What can a transformer express and learn?

3

LLMs in practice

How well do LLMs learn a language? What other phenomena do they exhibit?

abstract

concrete

Axis 1: No Assumption about the Computational Model

From positive examples alone, can *any* learner learn a language?

- Can we identify a language in the limit?
- Can we generate new strings from a language in the limit?

Axis 2: A Simplified Model of the Transformer

What is the *computational model* of a transformer?

- What programs can a transformer express?
- Is expressivity a good proxy for learnability?
- If a program is simple, does the transformer learn it?
- Can we derive an interpretable program from the transformer?

Axis 3: LLMs in Practice

How well do LLMs learn a language?
What other phenomena do they exhibit beyond learning?

- What languages are difficult to learn?
- What is the implicit world model of an LLM when learning?
- How do different modes of learning in LLMs compare?
- How does learning relate to memorization?

Logistics

Website	https://courses.mpi-sws.org/fllm-ws26
Class	Weekly, Thursday, 2–3:30 pm, Room 112 (MPI-SWS)
Office hours	Weekly, Friday, 2–3:30 pm, Room 112 (MPI-SWS)
Email	fllm26instructors@mpi-sws.org
Report submission	Via OLAT or Moodle — details to follow in the coming days
Paper selection	Submit by 23 October

Grading



- **Presentation.** 45–60 minutes on your paper, then Q&A
- **Report.** At most 2 pages, due before the next class; best three count
- **Participation.** Engagement in the Q&A

Passing

A combined score of **at least 50%**

What We Expect

A good presentation

- Motivate the question and formally state it
- Go deep on one proof or experiment; explain why you find it important
- Explain your takeaways from the paper

A good report on the presented paper

- Discuss at least one aspect of the paper that an AI reviewer, the presenter, or the original authors have missed
- Introduce insights to motivate future research

Class participation

Read the paper **before** the session and ask questions — the more difficult, the better

Syllabus

Date	Topic	Reading
22 Oct	Short intro and Q&A	—
29 Oct	No class	—
Axis 1 — No assumption about the computational model		
5 Nov	Generation in the Limit	Kleinberg & Mullainathan, *Language Generation in the Limit*, NeurIPS 2024
Axis 2 — A simplified model of the transformer		
12 Nov	Model of a Transformer	Weiss, Goldberg & Yahav, *Thinking Like Transformers*, ICML 2021
19 Nov	Predicting Length Generalization	Zhou et al., *What Algorithms Can Transformers Learn? A Study in Length Generalization*, ICLR 2024
26 Nov	Recovering the Learned Program	Friedman, Wettig & Chen, *Learning Transformer Programs*, NeurIPS 2023
Axis 3 — LLMs in practice		
3 Dec	Difficulty of Learning	Liu et al., *Exposing Attention Glitches with Flip-Flop Language Modeling*, NeurIPS 2023
14 Jan	Implicit World Models	Vafa et al., *Evaluating the World Model Implicit in a Generative Model*, NeurIPS 2024
21 Jan	Learning Modes	Ghosh et al., *Fine-Tuning vs. In-Context Learning in Large Language Models: A Formal Language Learning Perspective*, ACL 2026
28 Jan	Memorization & Learning	Ghosh et al., *Understanding the Interplay between Memorization and Learning in Large Language Models*, NeurIPS 2026

Next Steps



Before you leave today

Select at least three papers you would be happy to present, by 23 October

We assign papers on 26 October

Questions?



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